Lab for cluster randomized control trials

Case Study

Elementary School Project-Based Learning Program

Over the past five years, Dr. Wanda Smith and colleagues have been working closely with Detroit public schools to develop an innovative, hands-on science program for elementary school teachers and their students. During this time, Dr. Smith has worked closely with teachers in three elementary schools to develop and refine a professional development program and science units focused on environmental science and biology with children in grades 3-5. These units are hands-on and project-based and focus not only on scientific knowledge but also on centering students as scientists and observers in their communities. Over time, the team has developed five science units, associated activities, and teaching guidelines. Teachers are deeply engaged in this work and observations in the classrooms suggest that students are, indeed, making connections between science and their lives. Student knowledge of science seems to have increased, as does student motivation.

Plausible Evaluation Design

Clustered Random Assignment

Suppose researchers wish to evaluate the effectiveness of a similar program on student science motivation. Researchers enroll schools in the study during the summer and a portion of schools are randomized into the program while the balance of schools are employed as control clusters. Students who are enrolled into the study have their science learning motivation measured with a well known validated scale. A set of schools are recruited for the study from a variety of districts through a professional development conference for state principals and district membership is not considered in the design. For planning purposes, researchers use a school-level intraclass correlation of .05 based on a plan to match schools.

The analysis will employ a two-level hierarchical linear model (students nested within schools) with a random intercept to compare the averages of the treatment and control students.

For simplicity, assume perfect compliance with random assignment and a perfect response rate.

Question 1

Using the planner tool, find the minimum detectable effect for a `Clustered Random Assignment` study in which 6 schools are in the treatment group and 6 schools are in the control group, in each of which 20 students are surveyed for the computation of the motivation scale, when power is .8 for a test with a significance level of .05.

Question 2

The total sample in Question 1 is 240. A member of the study team claims that a better study would use 5 schools in treatment and 5 schools in control and 50 students in each school for a total sample of 500, arguing that doubling the sample would vastly increase the sensitivity.

Using the planner tool, find the minimum detectable effect for a `Clustered Random Assignment` study in which 5 schools are in the treatment group and 5 schools are in the control group, in each of which 50 students are surveyed for the computation of the motivation scale, when power is .8 for a test with a significance level of .05.

Is this design much more sensitive?

Question 3

Another researcher claims that instead of doubling the sample, the same resources can be used to pre-test the students in each school. Earlier analyses of the motivation instrument show that correlation between surveys is about $\rho = .9$ at the student level (for an R-square value of $\rho^2 = .81$) and about $\rho = .3$ at the school level (for an R-square value of $\rho^2 = .09$).

Using the planner tool, find the minimum detectable effect for a `Clustered Random Assignment` study in which 6 schools are in the treatment group and 6 schools are in the control group, in each of which 20 students are surveyed for the computation of the motivation scale, when power is .8 for a test with a significance level of .05.

Assuming that the strategy for Question 2 is costs the same amount as this strategy, which produces the more sensitive study?

Question 4

Another researcher claims that instead of using a pretest, the same resources can be used to gather data in more comparison schools. Doing so may increase the ICC to .1, but could yield 12 comparison schools in total.

Using the planner tool, find the minimum detectable effect for a `Clustered Random Assignment` study in which 6 schools are in the treatment group and 12 schools are in the control group, in each of which 20 students are surveyed for the computation of the motivation scale, but with an ICC of .1, when power is .8 for a test with a significance level of .05.

Do you think this is a good idea?

References

Barron,B. J., Schwartz, D. L., Vye, N. J., Moore, A., Petrosino, A., Zech, L., & Bransford, J. D. (1998). Doing with understanding: Lessons from research on problem-and project-based learning. *Journal of the learning sciences*, *7*(3-4), 271-311.